

their noise-free control in all weather conditions, along with absolute safety and reliability. With flying cars, passengers can freely enjoy the scenery from the sky, and go to places and avoid traffic jams.

A flying car is a general term for mobility with electric, autopilot, and vertical take-off and landing functions. Electrical vertical take-off and landing (eVTOL) is another term used to describe this technology in the urban air mobility sector. Uber has announced that it will start a business as a service (BaaS) platform in 2023. This gives eVTOL high hopes to become a key part of mobility as a service (MaaS).

Governments from different countries are working on public-private ventures for future air mobility, and a roadmap for land, infrastructure, transport and tourism is on its way for establishing the same. Air mobility has been proposed to be used as a means of alleviation of congestion in devel-

oped countries, emergency use and transportation, which does not require infrastructure in developing countries.

SkyDrive Inc. has been established by members of Cartivator, which is a group of aircraft, drones and automotive engineers for the development, manufacture and sales of flying cars. It successfully completed flight testing in 2019 in a joint effort with Toyota. They are planning to start manufacturing in 2020, and will be ready for public sales in 2023.

The first Japanese outdoor flight testing of a flying car (unmanned) was conducted with regular testing throughout 2019. Tomohiro Fukuzawa, chief executive, SkyDrive, says, "Flying cars will improve the lives of consumers across the world by providing efficiency of movement and creating happiness in flying—be it linking people to jobs, their family or a dream holiday. SkyDrive will be the travel mode of the future. The company is

now working on product development and certification."

SkyDrive has a length of 3600mm, width of 1700mm during travel and 3100mm during flight, and height of 1100mm. It can achieve maximum flight speed up to 100kmph and maximum driving speed up to 60kmph. It can drive up to flight altitude of 50m with a battery-operated engine. Its maximum take-off weight is 400kg.

Toyota City provides places for development and verification testing, discussions with country and prefecture about legislation, sorting issues for regional industrialisation, provision of opportunity for matching and so on. Cartivator and SkyDrive Inc. have taken charge of vehicle development, performance of flight and verification testing, provision of examination data, creation and improvement of the business model for industrialisation and so on.

—Deepshikha Shukla

4 AN AI COMPANION FOR KIDS

Traditionally, joint families have been a part of Indian culture. But in today's fast-growing society, disintegration of the joint family system has paved the way for nuclear families. It is no hidden fact that parents tend to spend less time with children due to their busy lifestyles. So how do children get the much-needed attention in their growing years and parents do not miss out on their lives?

Miko, a Mumbai-based robotics company founded by Prashant V. Iyengar, Sneh R. Vaswani and Chintan S. Raikar, aims to solve this challenge by applying artificial intelligence (AI), robotics and the Internet of Things (IoT) through its emotionally-intelligent solutions. The first product line, Miko series of personal AI robots, ensures that children undergo constant development and learning by educating and entertaining them in a responsible and fun way.

The latest product, Miko 2, stands 18cm tall and runs on 1.5GHz quad-core processor. It weighs 900 grams

and has a 7.1cm (2.8-inch)-wide, high-resolution IPS display. To ask a question, the child has to say "Hello, Miko" or press and hold the select button. The bot can hear the child even from a distance and estimate his/her direction to have a face-to-face conversation in English. It attempts to understand the child's emotions and improves with every interaction.

Kids can revise educational concepts, play games and listen to stories as well. With the latest update, the bot can be used to click pictures, too. A red light indicates that it needs charging. However, the bot should not be placed on glass surfaces.

Vaswani, co-founder and chief executive officer, Miko, says, "Compared to its predecessor, Miko 2 is smartphone-independent, Wi-Fi-enabled with advanced tech specifications and sensors, video-calling features, better microphones, front camera, and mood-, speech- and edge-detection abilities. Using state-of-the-art computer vision, the camera helps Miko



2 identify, remember and recollect known faces, objects and the world around it."

Miko 2 has an edge sensor that intelligently saves itself from falling from a surface while playing with children. It comes bundled with some of the best technologies globally that make the entire experience life-like. With over 34 sensor lines processed in real time, Miko 2 truly revolutionises man-machine interaction to the next level."

Safety is ensured by encrypting the child's data. Parents can remotely